

CVaR-Based Portfolio Optimisation for a Malta-Accessible UCITS ETF Basket

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Abstract—This paper offers some additional guidelines for MCAST IICT 2nd year B.Sc. students on research paper writing. The abstract is the first paragraph that any researcher reads and thus you need to capture the essence of your research. Dedicate 1 sentence for the theme (subject matter), another for your project aim, one for the proposed solution, then for the general outcome of this project (positive/negative result/outcome), another for the technology used (Windows/Linux, Android/iOS, C#/Java, Unity/Unreal), finally for any technique used (Neural Networks, Pearl Noise, Market Basket Analysis, K-Means, HMM, Augmented Reality). Following is an example: *In this research we are tackling the automatic annotation of cast members and use of key props within movies. Movies tend to gather a huge fan gathering who dedicate a lot of time and resources for documenting the content, thus tools to automate processes such as the timeline when cast members appear, presents of props such as weapons and other forms of annotations are desired. In this research we propose the use of image processing techniques, namely Principal Component Analysis (PCA) and Convolutional Neural Networks (CNN) for the automatic annotation of actors' timeline and presence of weapons in the popular Tv series Game of Thrones (GoT). Our proposed solution managed to detect the targets in 85% of the cases and we identify situations where this is challenging and recommend future research directions.*

Index Terms—MCAST, IICT, L^AT_EX, Project, Paper

I. INTRODUCTION

Recent market crises, including the COVID-19 shock, have renewed criticism of classical portfolio theory. Although the mean-variance (MV) framework remains foundational, variance treats upside and downside deviations symmetrically and therefore does not isolate severe losses in the way downside-focused investors typically require [1], [2]. For this reason, portfolio research has increasingly adopted Conditional Value-at-Risk (CVaR), which estimates average losses beyond the VaR threshold and can be incorporated into tractable optimisation formulations [2], [3]. Exchange-traded funds (ETFs) provide a practical setting for such comparison because prior studies have evaluated CVaR-based, MV, and equal-weight approaches using diversified ETF universes and out-of-sample backtesting [3].

Despite this relevance, much of the recent CVaR literature is embedded in advanced settings such as evolutionary algorithms, copula-based dependence modelling, bilevel multi-market optimisation, deep reinforcement learning, and distributionally robust optimisation [2]–[6]. These studies are

valuable, but they make it harder to judge whether a simpler and more transparent CVaR-minimisation framework, implemented under standard long-only constraints, already provides meaningful downside-risk benefits. In addition, while Malta has an established collective investment scheme environment and the Maltese Assets Fund illustrates the local relevance of diversified fund products [7], [8], there is limited evidence on CVaR-based optimisation in a Malta-accessible UCITS ETF context. The present study is therefore positioned at the intersection of quantitative portfolio theory and applied investment management, addressing a contextual and empirical gap by testing whether a Malta-accessible UCITS ETF basket can serve as a credible benchmark setting for comparing CVaR, Equal-Weight, and Minimum-Variance strategies.

The purpose of this quantitative backtesting study is to evaluate whether CVaR-minimisation provides stronger downside-risk control than Equal-Weight (1/N) and Minimum-Variance portfolio construction in a Malta-accessible UCITS ETF basket over 2016–2025. The independent variable is the optimisation strategy, operationalised through three portfolio rules: Equal-Weight, Minimum-Variance, and CVaR-minimisation. The dependent variables are the observed portfolio outcomes, namely CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown. To preserve comparability, the ETF universe, estimation window length, rebalancing frequency, and long-only fully invested constraints are treated as control variables. The study aims to implement the three strategies under identical conditions using rolling out-of-sample backtesting and to compare their performance on both return-based and downside-risk criteria.

Accordingly, the main hypothesis states that a CVaR-minimised portfolio will achieve lower CVaR and lower maximum drawdown than Equal-Weight and Minimum-Variance portfolios without material deterioration in CAGR or Sharpe ratio. The null hypothesis states that no meaningful difference will exist among the three strategies in CVaR, maximum drawdown, or risk-adjusted performance. As shown in Fig. 1, the present study adopts a positivist philosophy and a deductive approach. It employs a controlled experimental backtesting strategy using secondary historical ETF data, a mono-method quantitative design, and a longitudinal time horizon. The techniques and procedures consist of secondary ETF data

collection, rolling out-of-sample backtesting, and quantitative performance and downside-risk analysis. Specifically, portfolio outcomes are treated as objectively measurable through predefined financial metrics. A deductive approach is adopted because the study begins from established theory on the limitations of variance as a risk proxy and tests whether a downside-risk measure produces superior empirical outcomes [1], [2]. Since performance is evaluated across repeated windows over 2016–2025 rather than at a single point in time, the time horizon is longitudinal.

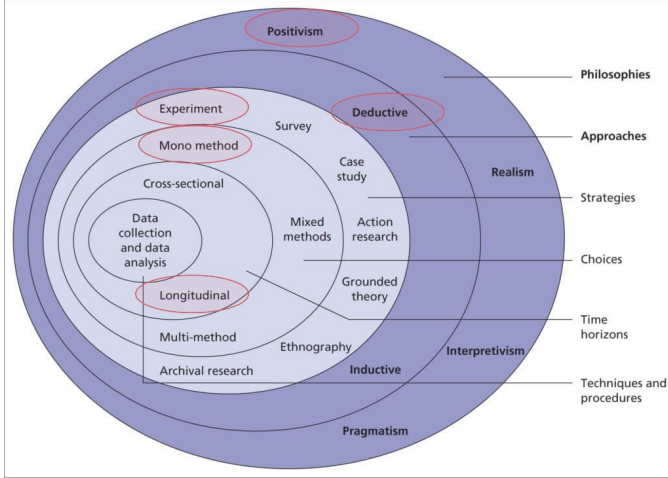


Fig. 1. Research onion for the present study, adapted from Fig. 4.1 in Saunders, Lewis, and Thornhill [9].

II. LITERATURE REVIEW

Section II reviews the research methodologies used in six peer-reviewed journal articles [1]– [6]. Non-academic and institutional sources are used only for contextual background and are excluded from the methodological comparison.

Fig. 2 organises the reviewed methodological literature into data employed, process and candidate techniques, and evaluation and testing strategies. The map shows that prior studies mainly rely on secondary financial data and backtesting-based evaluation, but differ substantially in portfolio-construction complexity. The final branch highlights the gap addressed by the present study, namely a simpler and more transparent comparison of Equal-Weight, Minimum-Variance, and CVaR-minimisation strategies in a Malta-accessible UCITS ETF setting under identical long-only conditions.

A. Overview of reviewed methodologies

A consistent methodological pattern across the reviewed studies is the use of a positivist, quantitative design based on secondary financial data. Rather than using interviews or surveys, the reviewed papers typically rely on historical price or return series and assess portfolio-construction rules through computational testing. This broad methodological direction aligns closely with the present study, which also examines objectively measurable portfolio outcomes under controlled backtesting conditions. Accordingly, the literature map for the

present study is organised around data, process, and evaluation, as illustrated in Fig. 2.

B. Data employed in prior studies

The reviewed literature shows that portfolio-optimisation research commonly uses historical market data covering several years so that both normal and stressed market conditions are represented. Muñoz and Cifuentes [1], for example, analyse historical returns from 2004 to 2023 and deliberately include major crisis periods to test the stability of mean-variance and CVaR formulations. Their use of rolling windows is methodologically valuable because it allows repeated comparison across time rather than relying on a single in-sample period.

Teplova et al. [3] also use a long historical dataset, drawing on daily returns from international ETFs between 2012 and 2022. This is particularly relevant because ETFs are also the asset class used in the present study. Their data design supports realistic comparison between portfolio rules while also enabling out-of-sample testing.

By contrast, Nasini et al. [4] use a much larger firm-level dataset across thousands of U.S. listed companies, while Cui et al. [5] focus on cryptocurrency returns. Kobayashi et al. [6] also work with large asset universes in a robust optimisation setting. These studies show that broader datasets are possible, but they also increase modelling and computational complexity. For the current research, a Malta-accessible UCITS ETF basket is more appropriate because it remains sufficiently diversified while preserving transparency, feasibility, and direct relevance to the stated research context.

Overall, the literature suggests that the most suitable data strategy for the present study is a secondary dataset of historical ETF prices and returns, covering a sufficiently long period to include both calm and turbulent market conditions, while keeping the investable universe constant across all tested strategies.

C. Portfolio-construction process and candidate techniques

The reviewed studies show that the main methodological distinction lies in the portfolio-construction process. Some studies compare classical benchmark approaches, while others embed CVaR within more advanced optimisation frameworks.

Muñoz and Cifuentes [1] provide the clearest benchmark-comparison methodology. They compare mean-variance and CVaR formulations under the same target-return structure, which isolates the effect of the risk metric itself. This is highly relevant to the present study because it demonstrates how a fair experimental comparison can be achieved when constraints and data inputs are held constant.

Teplova et al. [3] extend the process stage through copula modelling and a Black-Litterman mean-CVaR framework. Their methodology is rigorous, but it depends on parameter calibration, investor-view modelling, and dependence-structure estimation. Kaucic et al. [2] instead treat the process stage as a multi-objective evolutionary search problem, comparing variants of NSGA-II and SPEA 2 under downside-risk measures such as semi-variance and CVaR. Nasini et al. [4] introduce

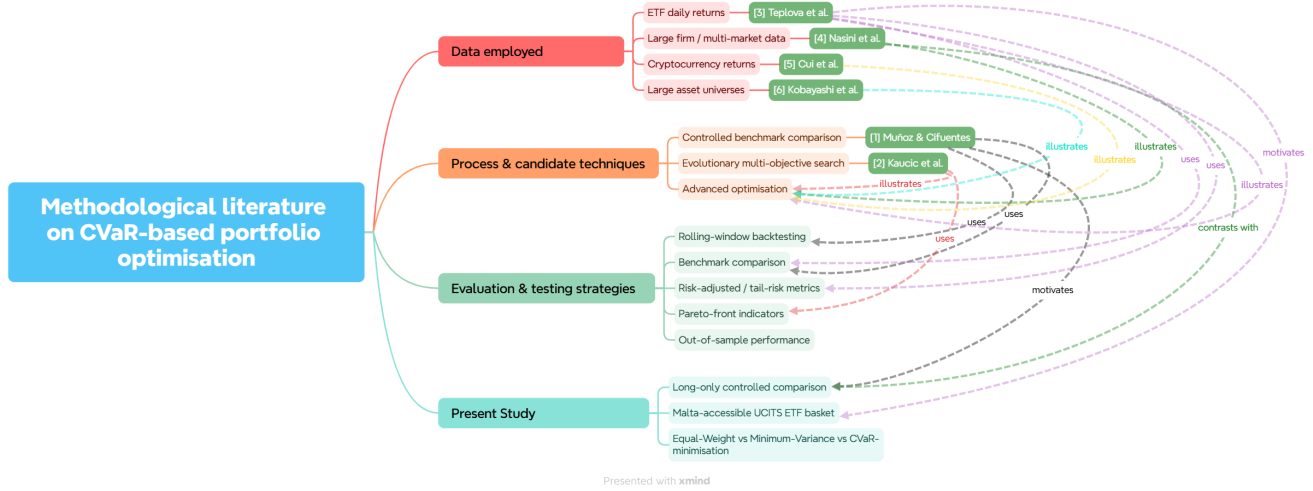


Fig. 2. Methodological literature map for CVaR-based portfolio optimisation.

a bilevel multi-market formulation, while Kobayashi et al. [6] use distributionally robust optimisation with cardinality constraints. Cui et al. [5] move further still by integrating deep reinforcement learning into a mean-CVaR framework.

These advanced methodologies are useful as evidence that CVaR is a mature and flexible risk measure, but they are not ideal templates for the present project. The current study is not intended to propose a highly specialised optimisation engine. Instead, it aims to compare three portfolio-construction rules under identical conditions: Equal-Weight, Minimum-Variance, and CVaR-minimisation. This choice is methodologically stronger for the present study because it prioritises interpretability, control, and reproducibility over algorithmic novelty.

D. Evaluation and testing strategies

The literature also shows a clear methodological pattern in the evaluation stage. Studies rarely stop at in-sample optimisation. Instead, they use out-of-sample testing and compare strategies through financial performance and risk metrics.

Muñoz and Cifuentes [1] use rolling windows and assess the similarity or difference between resulting portfolios under different risk formulations. Teplova et al. [3] evaluate portfolio performance against benchmark strategies and report risk-adjusted outcomes as well as tail-risk behaviour. Kobayashi et al. [6] also use out-of-sample investment performance measures, while Kaucic et al. [2] assess optimisation quality through Pareto-front indicators in addition to portfolio risk measures.

Across the reviewed studies, the most relevant evaluation measures for the present research are those that capture both return and downside risk. In particular, CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum draw-down appear suitable because they allow direct comparison between standard performance and tail-risk protection. This is important because the study hypothesis does not claim

that CVaR-minimisation will maximise returns; rather, it tests whether CVaR-minimisation improves downside-risk control without materially worsening return-based outcomes.

E. Comparative synthesis and methodological gap

Taken together, the reviewed literature supports three main conclusions. First, the dominant research tradition in this area is quantitative, secondary-data, and backtesting-based. This strongly supports the philosophical and methodological position already established in Section I and illustrated in Fig. 1.

Second, the strongest comparative designs are those that keep the dataset, constraints, and testing framework constant while varying only the portfolio rule. This is the methodological logic most clearly visible in Muñoz and Cifuentes [1], and it is the one most appropriate for the present study.

Third, although advanced methods such as copulas, bilevel programming, evolutionary optimisation, robust optimisation, and deep reinforcement learning increase technical novelty, they also introduce greater complexity and reduce transparency. For this research a simpler controlled comparison of benchmark strategies is methodologically more appropriate.

Finally, across the reviewed studies, the main independent element is the portfolio-construction or optimisation method, while the dependent outcomes are portfolio performance and downside-risk measures assessed through out-of-sample testing. Common testing strategies include rolling-window backtesting, benchmark comparison, and risk-adjusted performance evaluation. However, despite this methodological maturity, limited evidence compares CVaR-minimisation, minimum-variance, and equal-weight strategies in a simple, transparent, Malta-accessible UCITS ETF setting under identical long-only backtesting conditions.

III. RESEARCH METHODOLOGY

This section is one of the most important parts of your research and it is the one that will receive most criticism. The reason is that here you document how you conducted your research. Since you are reading a B.Sc. degree you are expected to follow a scientific method, thus in this section you are expected to document: **the problem; hypothesis; aim & objectives; research questions; and research pipeline.** Let's go through each.

First start by providing a brief overview of the problem, similar to what you already mentioned in the previous sections. Then provide a hypothesis, this is the assumption you are making, which via your research you are attempting to prove or disprove. An example hypothesis in traffic management would be: *By using image processing on public camera video feeds we believe that it is possible to estimate the traffic congestion at a higher precision than public traffic data feeds.*

After explaining the problem and hypothesis you need to specify why you are doing this research and what you think the output will be, so the aim & objectives. Let us consider the "Marine Vessel Tracking" concept: *With this research we aim to be able to automatically detect any anomalies within Marine traffic, such as drug smuggling, human trafficking and/or collisions.* Our objectives are:

- 1) Identify existing data sets and manners of connecting them or even enriching them.
- 2) Review existing algorithms and propose a suitable model
- 3) Evaluate the effectiveness of the proposed model with real data and existing research.

So lastly you have to present a number of research questions that you attempt to answer, which ideally revolve around the **data set, algorithm** and **evaluation** which you documented in your literature review:

- 1) Does a data set need to be created for such a research?
- 2) Which current technology is best suited to address the problem?
- 3) How does the proposed solution compare with existing solutions?

Finally, you need a plan on how you intend to answer the research questions which will help you prove/disprove the hypothesis. Different research areas would require specific pipelines, yet a generic one is shown in Fig 3. After presenting an illustration of your pipeline it is recommended to document what has been done at every stage. Remember you should use past tense in this section since you are document what you have already done, since this paper will be published at the end of your research.

IV. FINDINGS & DISCUSSION OF RESULTS

This section is a delicate section and you are encouraged to be as honest and open as possible. The aim is not to show that you got a perfect solution to a long existing problem. Trying to state that in a few months you developed a perfect solution with 100% is not convincing and raises doubts. It is recommended that you document your findings in every

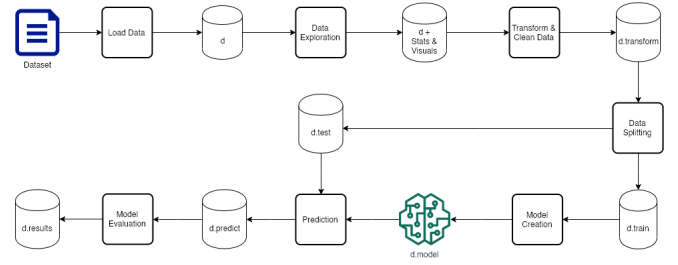


Fig. 3. Research Pipeline

step of your research pipeline, highlighting your observations and decisions taken. Present your results and very importantly compare with existing research which you documented in Section II.

The focus of the Project module is for you to delve into an area that exposes you to new technologies and offers you an opportunity to be critical of your work. So you are expected to document where your solution/research worked and where it did not. Reflect and document reasons why the solution/research did not perform as expected and propose ways of addressing this. From these observations you will produce new research questions in the next section. Consider the research in "Brand usage detection within audio streams", where certain key terms were searched within videos, the results of which are documented in Table I. You will notice that the terms "Peppa" and "Sushi" were the least recognised terms even by the best transcribers. Upon investigation we determined that "Peppa" was not recognised cause of voice morphing to create childish voices in the cartoon video, whilst "Sushi" was pronounced by a Japanese person speaking English. So the research student decided to focus his dissertation research on how to create a system that is able to recognise heavy accents to automate the configuration of a transcriber, in this case to cater for English spoken by a Japanese person, which accent is very different from an Indian accent, British accent or Italian accent, just to name a few.

TABLE I
RECALL RESULTS

#	Term	Google Cloud	Google Speech	Sphinx CMU
1	Peppa	27%	33%	0%
2	Peppa	33%	22%	0%
3	Apple	96%	92%	79%
4	Galaxy	100%	100%	100%
5	Galaxy	95%	95%	80%
6	Sushi	75%	35%	0%
	Average	71%	62%	43%

V. CONCLUSION

So the conclusion is most probably the second section that a reader would use to consider reading in full your research. Thus it is important to highlight the essence of your research. The recommended approach is to answer your research methodology. Start by answering your research questions, then stating to what degree did this research achieve its aim and

objectives, highlighting potential causes for not being able to do so at a desired level, such as time, or other circumstances. Consider the following: *A student was due to research the use of MCAST computers during out-of-office hours to offer a private cloud computing service for research, similar to Google Cloud but free for MCAST students. Due to the lock-down by COVID-19 pandemic we could not continue on the original planned research objective and had to adapt.*

The final and most important part of the conclusion are your recommendations for future research, not necessarily for yourself (referring to what you plan to do in your dissertation or beyond), but also to other future researchers who might consider doing similar work to yours. The recommendations you provide here will set such prospective researchers on a better track/direction thanks to your experience.

APPENDIX A SUPPORTING MATERIAL

You can add screen shots and statistics. Stick to essential information.

ACKNOWLEDGEMENT

You can dedicate this section for special assistance that you were given by a 3rd party. Not your mentor, relatives or questionnaire participants.

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